

PRATIK GANESH MESHARAM

Machine Learning & IoT Systems Engineer

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SUMMARY

Computer Science Graduate (B.Tech, VIT Pune, CGPA 8.33) building and shipping end-to-end machine learning systems — from data pipelines and model training through deployment in web and mobile interfaces. National winner, Smart India Hackathon 2025, for a computer-vision plankton classification system. Comfortable across the stack: Python/R model development, Scikit-learn and YOLOv8 pipelines, and Flask/React deployment.

TECHNICAL SKILLS

Languages: Python, R, C/C++, SQL

Machine Learning: Scikit-learn, Random Forest, Logistic Regression, SVM, Naive Bayes, Decision Trees, Boruta Feature Selection, Ensemble Model Comparison

Deep Learning / Computer Vision: YOLOv8, MobileNetV3, Convolutional Neural Networks (CNN), Artificial Neural Networks (ANN), OpenCV, Ultralytics.

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, R Shiny, Exploratory Data Analysis

IoT & Embedded Systems: ESP32, Arduino, MQTT, Adafruit IO, Firebase/Firestore

Web & App Development: Flask, React, Vite, Tailwind CSS

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS code Vercel, Fast APIs

ACADEMIC PROJECTS

Plankton AI — Autonomous Plankton Detection System

2025

Smart India Hackathon (SIH) 2025 — 1st Place, National Winner | Python, YOLOv8n, MobileNetV3, OpenCV

- Designed and implemented a computer-vision pipeline for automated plankton detection and species classification from microscopic imagery, covering preprocessing, augmentation, and feature extraction.
- Built and compared object-detection and classification models using YOLOv8n and MobileNetV3, benchmarking on accuracy, precision, recall, and F1-score to select the deployment-ready model.
- Delivered an end-to-end pipeline that took the project from prototype to a nationally recognized, judged solution among hundreds of teams.

Aero Flame — Fire Surveillance Drone

2024

ESP32, OpenCV, BetaFlight, Flask, Ultralytics YOLO

- Built a real-time fire and flame detection system using an Ultralytics YOLO model, processing each video frame through OpenCV to draw bounding boxes and confidence scores around detections.
- Integrated the detection pipeline into a Flask web application supporting live video upload and streaming inference, enabling automated fire alerts through the web interface.

Lung Cancer Risk Prediction — ML Pipeline

2023

R, Boruta, Random Forest, Logistic Regression, Shiny

- Built a complete predictive-modeling pipeline in R for lung cancer risk assessment, from data cleaning and feature engineering through model evaluation and deployment.
- Applied Boruta feature selection to isolate the most predictive clinical and lifestyle variables, then trained and compared Random Forest and Logistic Regression models to identify the strongest-performing classifier.
- Deployed the final model as an interactive R Shiny application, letting users input health indicators and receive a real-time risk assessment.

ACHIEVEMENTS & CERTIFICATIONS

- Smart India Hackathon (SIH) 2025 — 1st Place, National Winner, Plankton AI Autonomous Detection System.
- Certification: Python for Data Science, AI & Development — IBM, Fundamentals of Deep Learning — Nvidia, Decode DSA - PW Skills, Ethical Hacking — Udemy.

EDUCATION

Bachelor of Technology in Computer Science & Engineering — Vishwakarma Institute of Technology, Pune

2022 – 2026 | CGPA: 8.33 | SGPA : 9.8

Higher Secondary (Class XII) — Nowrosjee Wadia College, Pune | 2021 – 2022

Secondary School (Class X) — Dr. Mrs. Erin N. Nagarvala Day School, Pune | 2019 – 2020